Tech For Good 2024(Wellbeing) Proposal

Team Name: Wasteless Water

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Part 1: Introduction

Ocean pollution comes in many forms, but the all-time issue is with plastic pollution. While plastic has many substantial uses in our daily lives, we have become addicted to plastic products which causes severe environmental consequences.

The Earth's oceans, which encompass approximately 71% of its surface, are crucial for sustaining life, regulating the climate, and offering essential resources. Nevertheless, the continuous release of pollutants into these oceans presents substantial risks to marine life, ecosystems, and human well-being.

Every year, we produce 350 million tons of plastic which is equivalent to 58 million African Elephants. Half of the world’s plastic were manufactured for single use purposes. The rate we are using the plastics are the cause of the plastic waste to accumulate to such degrees.There are organisations, movements to stop plastic waste from its roots, but the accumulated plastic waste in the ocean is yet to be solved. Although actions have been made to make ocean clean again, but we aim to accelerate the cleaning process to stop further harm on the marine ecosystems.

Thus, we are introducing a system that can be easily installed into a ship to collect ocean waste while on the run. With this we could increase the efficiency and decrease the cost for sweeping up ocean waste.

Part 2: Introduction

The aim of this project is to control and cease ocean pollution from contamination waste produced by humans for the past 30 years. This project is to clean up the coastlines passively with the ongoing boats.

There is still no fixed cost to ensure the possibility of this project and the level of difficulty is immeasurable. It is calculated there are 290 thousand vessels operating at 5-6 hours every day, with about 60%-70% around the areas of plastic emissions.

In addressing this issue of ocean pollution, we can contribute to the achievement of several Sustainable and Development Goals (SDGs). Specifically:

- Life Below Water (Goal 14): By reducing ocean pollution, we contribute to the conservation and sustainable use of oceans, seas, and marine resources. This includes efforts to prevent marine pollution, protect marine ecosystems, and promote sustainable fisheries management.
- Clean Water and Sanitation (Goal 6): Ocean pollution often originates from land-based sources, such as industrial discharges and agricultural runoff. By mitigating these sources of pollution, we can improve water quality and contribute to achieving the goal of clean water and sanitation for all.
- Life on Land (Goal 15): Many land-based activities contribute to ocean pollution, affecting both terrestrial and marine ecosystems. Addressing ocean pollution can help protect terrestrial environments and halt biodiversity loss, contributing to the goal of life on land.

Part 3.1: Aim

As of the 21st century, there have already been many cases of pollutants being accidentally released into the water environment. Cases such as the explosion of the Deepwater Horizon in 2010, where an offshore oil rig in the Gulf of Mexico exploded due to structural safety standards negligence, had leaked over 4 million barrels of oil over 87 days, severely damaging the flora and fauna surrounding that area. It had been the largest marine oil spill in the history of the petroleum industry and consequently, the largest environmental disaster to ever occur in world history, with an estimated clean-up cost of up to $63.4 billion. With the current clean-up strategies used, it is not only expensive to scale, but even more so when it comes to the efficiency and time required to control the expansion of pollutants before it gets out of control. There are about 1.3 large-scale oil spills in the year 2020, with some estimates of up to 1.5 major oil spills happening per year, it is just not cost-effective nor is it viable in the long term to deal with pollutants with current filtration/containment methods, and so our team wants to find a better way to automate this clean-up process while at the same time making it easier to scale up to deal with larger scale environmental disasters.

Other than ocean contamination due to crude oil spillage, chemical spoilage into natural water bodies is indeed another problem of the status quo that is desperate for a solution. For example the Minamata Disease. It refers to a mass chemical waste disposal in Minamata, Japan. This event occurred in 1908 and has continued for around 36 years, and is responsible for around two thousand victims. Implementing filters at the factory's sewage works is easy, but cleaning up the contamination for those who fail to do so is hard as it requires lots of work and takes up a lot of time. This further hinders the process of cleaning up the chemical waste.

Our project relies on the traffic of the boats for the efficiency of cleaning up floatable wastes and chemical wastes such as crude oil and heavy metal . The increase of awareness of climate change and pollution in peoples had many organizations built to serve the purpose. But contamination still flows to the heart of the ocean every second. "Nearly one-quarter of the world's plastic waste is mismanaged or littered. Around 82 million tonnes.

One-quarter of that – 19 million tonnes – is leaked to the environment. 13 million tonnes to terrestrial environments, and 6 million tonnes to rivers or coastlines. 1.7 million tons of this are then transported to the ocean: 1.4 million tonnes from rivers, and 0.3 million tonnes from coastlines. The rest of the plastic waste that was leaked into aquatic environments accumulated in rivers and lakes." -Hannah Ritchie (2023) Studies shown that it is estimated that around 1000 rivers are accounted for nearly 80% of the annual riverine plastic emissions into the sea. A study comparison of marine traffic and riverine plastic emissions shows that its similarity is at around 90%, which supports the possibility of a boat filter for coastlines to cut off plastic waste.

In conclusion, water contamination is a pressing crisis that needs to be solved as soon as possible. However, there isn't a way to detoxify the contaminated water by using a single product. The current solution of using algae and enzymes to clean up oil spillage cannot be implemented into chemical waste spillage, and the solution to chemical spillage often ties with only compensation which provides no redemption to the natural water bodies. In the same time there isn't a product that can do both water detoxication and solid waste cleaning. Hence we propose another method that can be implemented in both scenarios.

Part 3.2: Objective

With water pollution at an all-time high, our aim is simple, to filter out as much polluted water as required by safety standards without inducing too much manpower and time, while simultaneously reducing the cost required to run the operation. With the current AI technology and filtering system (etc reverse osmosis) presented, our ways of filtering heavily revolved around using AI to automate the filtration, navigation and identification process of the ship, using reverse osmosis method to filter out all the pollutants such as heavy metals and oil spills, and finally releasing the clean water back into the environment while the filtered pollutants can be dealt with by either neutralizing them chemically, stored physically, or even recycled into usable products. The objective is to clear out polluted water, for example on large-scale oil spills or after a serious ship accident happened in an area, where hydrocarbons or heavy metals had leaked into the surrounding waters, to prevent the expansion of polluted areas, and revert the affected environment to its health and biodiversity. Or plastic and solid waste from daily rubbish that has been discarded into the water.

In short, we aim to develop a machine, more specifically a type of module that can be attached on boats that can filtrate water back to its natural state. Hence allowing nature water bodies to revert back to their original state. This will ensure the optimum living environment for the ecosystem in the water bodies, hence protecting the environment.

Part 4.1: Methods

The approach for the problem of the status quo will be a device that will be able to cleanse the water using reverse osmosis. The device with the name ____ will be a modular system that consist of three parts: The Container System, the Purification System and the Solid Waste System The purpose of the product is to cleanse the water that is being polluted by factory when they release the waste water into different water bodies around the globe or he crude oil spillage which causes great harm to the environment. And hence as such our primary goal will be removing the pollutant in the waste water, specifically heavy metal which causes the most harm to the environment, and crude oil particle from oil spillage.

The design of our product will be a few modular systems that can be attached on boats from small sampans to large oil tanks. The modular system includes the Carrier System that will be carrying the Filtration System when the container system consists of two parts, Cleaned Chamber and Waste Chamber. The purified water will then be stored in the Cleaned Chamber. The boat will then carry the water in the Cleaned Chamber into the places where the water is not contaminated and release the decontaminated water there. This is to prevent further recontamination from the unwanted waste. The filtered contaminated particles such as heavy metal and crude oil will be stored in Waste Chamber which we will further process it or discard it by other method that will not cause harm to the environment such as bacterial decomposing. We will also be adding a solid waste filtration system which will enables us to collect the solid waste that is present in the water bodies. And as such, we will be tackling the problem of contamination and pollution of factory waste water and crude oil spillage together with solid waste.

The basic principle of the design is to use reverse osmosis and ultrafiltration to eliminate the contaminants in the water such as heavy metals, crude oil particles and other particles that could possibly harm the environment with their presence in the water. For our design, we plan to solve the water contamination problem around the globe ranging from heavy metal pollution to oil tank spillage. After careful consideration, we have chosen two processes for this project. The first is ultrafiltration and the second one is reverse osmosis. The presence of reverse osmosis is by using a semi-permeable membrane to separate water molecules from other bigger particles (4.1 BYJU’s n.d.). This process is done by applying great pressure through the water with the presence of a semi-permeable membrane. The

contaminated water is forced to pass through the membrane, and as the semipermeable membrane only molecules with a size lesser or equal to water molecules pass through(4.2 T.Stricklin,2020), the contaminants which usually have a greater molecular size will be forced to retain on the other side of the membrane where the water molecule will be allowed to pass through the membrane (4.3Quench 2019).In order to achieve this effect, external pressure is applied on the side of the liquid with a high concentration of unwanted particles. This will allow the natural flow of water to be reversed (4.4 AquaCure, n.d.) Hence as such, clean water will pass through the membrane and hence filtration is achieved. Reverse osmosis is used to tackle crude oil spillage and waste metal contamination. A picture of detailed reverse osmosis is attached below

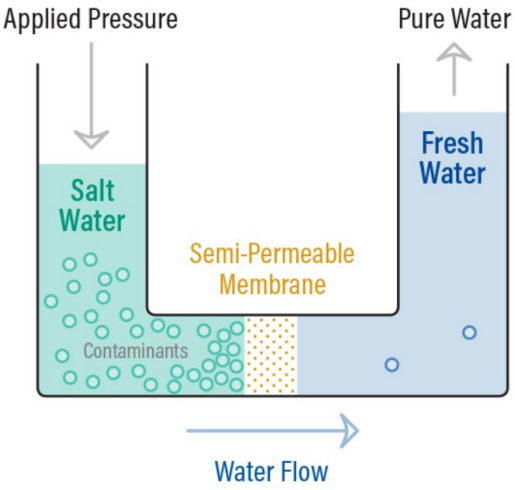


Figure 4.1 Process of reverse osmosis (4.5 Puretec Industrial Water, 2012)

Crude oil particle ranges from1.8 μm to 12.5 μm and has a mean diameter of around 6.25 μm(4.6 F. Wang, Z. Li, P. Wang, and R. Zhang 2018) and are usually insoluble in water due to the nature of petroleum being non-polar(4.7 Inai Martins Ribeiro de Andrade Bruning, n.d.) when water is a polar solvent(4.8 David Foster Wallace, 2005). According to the principle of dissolution, polar solvents will only be able to dissolve polar solvents (4.9 B. Zhuang, G. Ramanauskaite, Z. Y. Koa, and Z.-G. Wang, 2021). And hence petroleum will not be able to be dissolved in water. This will allow the petroleum particle will be kept at the other end of the semi-permeable membrane when the clean, purified water passes through the permeable membrane and into the clean-containing chamber. In the same time, heavy metal particles are also insoluble due to it's nature of being metallic molecule rather than being a polar molecule. This prevents it from being soluble.

However, to ensure that the water produced will be the cleanest and without the presence of crude oil or other possible contaminants that could possibly harm the environment, a pre-process is being implemented. Which is the ultrafiltration. Ultrafiltration refers to a process of water purification process that will be able to eliminate most of the unwanted particles through a mechanical filter which has an extremely fine filtration hole in it (4.10 F.N. Team, 2021). The nature of ultrafiltration is that it stops the unwanted contaminated particles before the filtration holes hence allowing the unwanted particles to be left over there. Ultrafiltration can filter particles of up to 0.1 to 0.001 μm. The nature of removing high molecular weight substances makes it a perfect pre-process to eliminate heavy metals that are being produced from factories that might cause poisoning of nature (4.11 G.K.Dhawan n.d.).

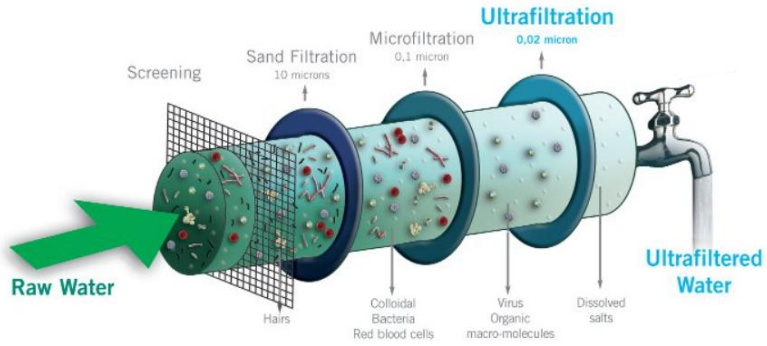


Figure 4.2: Process of ultrafiltration (4.12 Rekaindo, n.d.)

Ultrafiltration will be able to cleanse heavy metals out of the water. Heavy metal refers to a group of elements that have a high atomic mass or a density that is at least 5 times greater than water. (4.13 P. B. Tchounwou, C. G. Yedjou, A. K. Patlolla, and D. J. Sutton, 2014) and has the potential to lower energy levels and damage the functionality of the brain lungs kidneys liver and other vital organs. (4.14 M. Jaishankar, T. Tseten, N. Anbalagan, B. B. Mathew, and K. N. Beeregowda, 2014). Taking the Kuitun River in Xinjiang as a reference, the heavy metal usually ranges from 4 μm to 125 μm(4.15 D. He, X. Shi, and D. Wu, 2016). Hence as the ultrafiltration can filter up to 0.1 μm of particles it will be able to filter away most of the heavy metals hence minimize the damage done to the environment due to wastewater that is polluted into water bodies (4.16 D.-Q. Cao, 2020)

The reason for choosing two different processes is that ultrafiltration can ultrafiltration and reverse osmosis both provide different features which are important in purifying the water. Reverse osmosis can purify the water hence producing clean and safe water for the environment (4.17 Vedantu, n.d.). Whereas the ultrafiltration process can eliminate the heavy metals which protect the environment from heavy metal poisoning (4.18 N.W.S.P.Ltd, n.d.) Furthermore, the presence of ultrafiltration as the pre-treatment for reverse osmosis can protect the membrane of ultrafiltration and hence protect the membrane. This will significantly reduce the cleaning frequency and lower the cost of maintenance. (4.19 Dupont, n.d.). It can also lessen the membranes used in the reverse osmosis system, and hence eliminate one of the greatest disadvantages of reverse osmosis, which is the high set-up cost. (4.20 AO Smith Philippines, n.d.) And hence provide the best cost-effective design.

For the Solid Waste collection system, we plan to create a module that consist of two parts: The Filter and The Collector. The module will be placed in front of the boats. The Filter is a U- shaped space with nets at the end of it. The collector will be a robotic arm under The Filter which carry the solid waste away from the Filter. The Solid Waste collection process is illustrated below. Firstly, the gate of the Filter will be opened, the boat will move, and the water alongst with the solid waste will flow through The Filter. The waste which is solid in nature will be blocked and force to be left in The Filter when the water will be able to pass through The Filter. If the Water Purification Module was added, the water will be directed for further processing, A sensor will be placed at the end of the Filter, when the Solid Waste are cumulated to a certain level, the Collector Robotic Arm will lift and collect all the waste away from the U-Shaped Space in the Filter and in to the collecting bin. Which then will be compressed further to conserve space. By doing so we will be able to ensure that not only the chemicals, the solid waste can also be eliminated from water bodies

Part 4.2: Efficiency

There are two main parts in the Purification System: The Ultrafiltration device and the Reverse Osmosis device. We will be using pre-existing products for this project to ensure the most optimum and effective removal of waste. For the Ultrafiltration device, as the water we purify will not have to meet the drinking standard due to the presence of Reverse osmosis

process after ultrafiltration, we will be using point-of-entry ultrafiltration. We will be assuming that our project will be using the150mlh UF Membrane filter by Berghof Membranes (3.21 Berghof Membranes. N.d.). We will be using a total size of 2m square Ultrafiltration membrane filter which allows us to produce at least 300 litres of clean purified water per hour.

For the Reverse Osmosis System, we plan to use the Commercial Reverse Osmosis System 500lph RO Plant with Model Number JFRO-500L. It has a outlet flow of 500L/H. As the water purification speed of the Reverse Osmosis is faster than the water filtration speed of the Ultrafiltration device, the limiting factor that determines the efficiency of the water purification process will be the Ultrafiltration device. And hence the following calculation will be revolving around the effectiveness of the ultrafiltration device.

For the efficiency calculation, we will be taking the Penglai 19-3 Oil Spillage incident that happened in 2011 which causes around 4250 square kilometres of contaminated water bodies (3.22 AFP 2011) which is roughly 6 times the size of Singapore. It took months to resolve the oil spillage and the contamination done by it. But still the damage to the surrounding environment was beyond salvation. In our best-case scenario, we will be able to deal with the oil spillage in around 2 months. The calculation of the above-mentioned claim is attached as below

According to the description done by the Chinese Government, the colour of the oil film is between Silvery Sheen and having first traces of colour (3.23 J. Guo, X. Liu, and Q. Xie, 2012). Cross referencing with the oil spill volume chart, we can deduce that the total thickness of the film is between 76nm to 150nm. Taking the median between two values, we will be using 113nm of film thickness.

The total volume of the oil spillage will be $113 \times 10^{-9} \times 4250 = 0.00048025km^3 = 480250m^3 = 480250L$. Which will be taking around $\frac{480250}{300}$ =1600.83hours. Which is equivalent to 66days, approximately two months. Hence having a higher efficiency than the current status quo of several months. And hence our product will be able to provide an edge to the problem in the status quo.

Part 4.3: Design

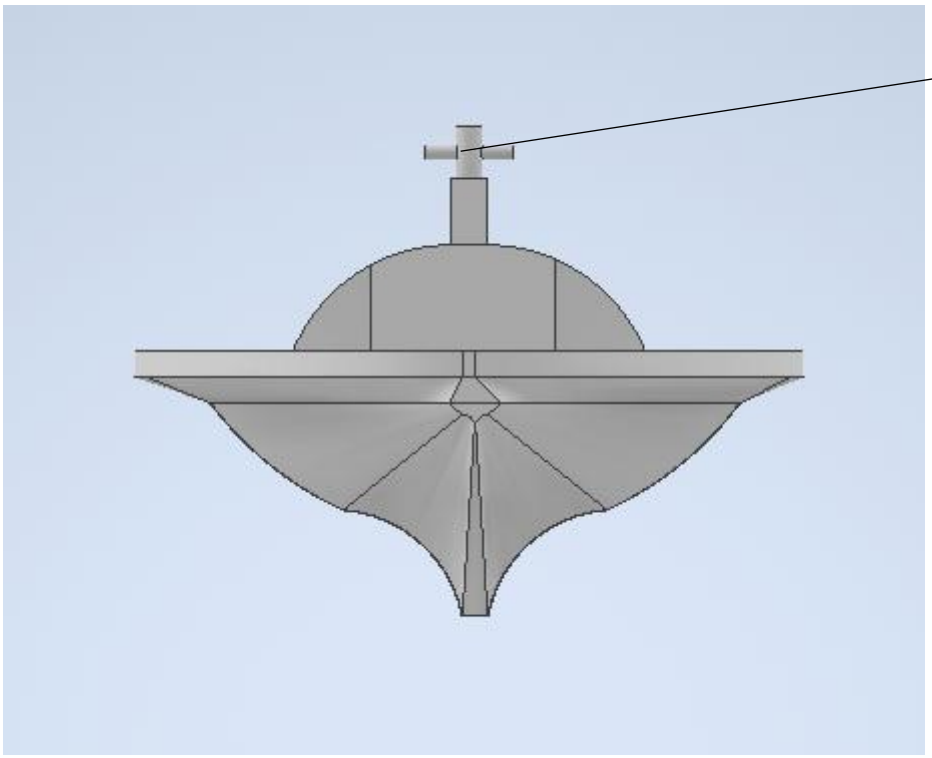


Diagram 3c Frontal View

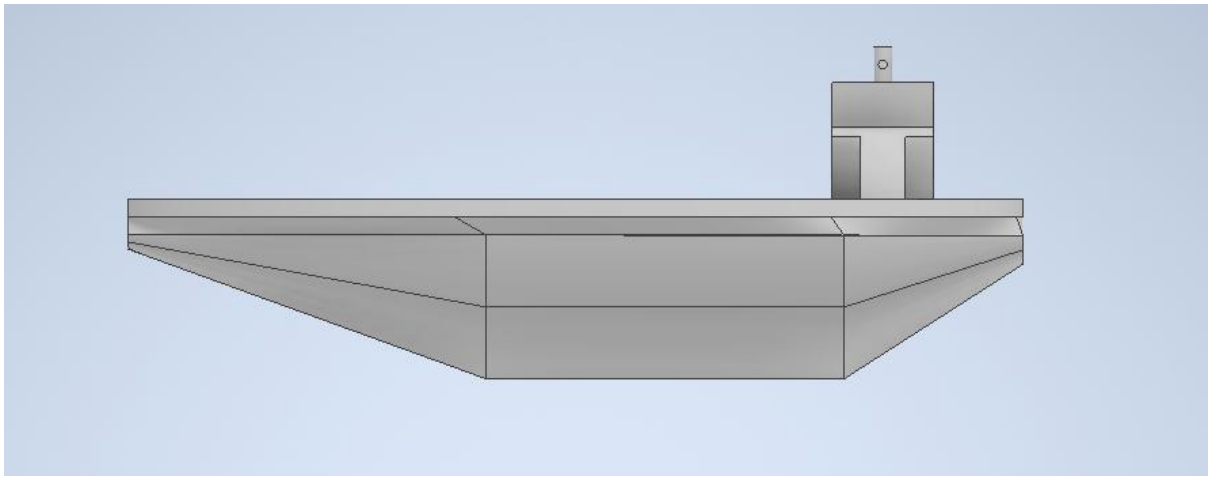


Diagram 3d Side View

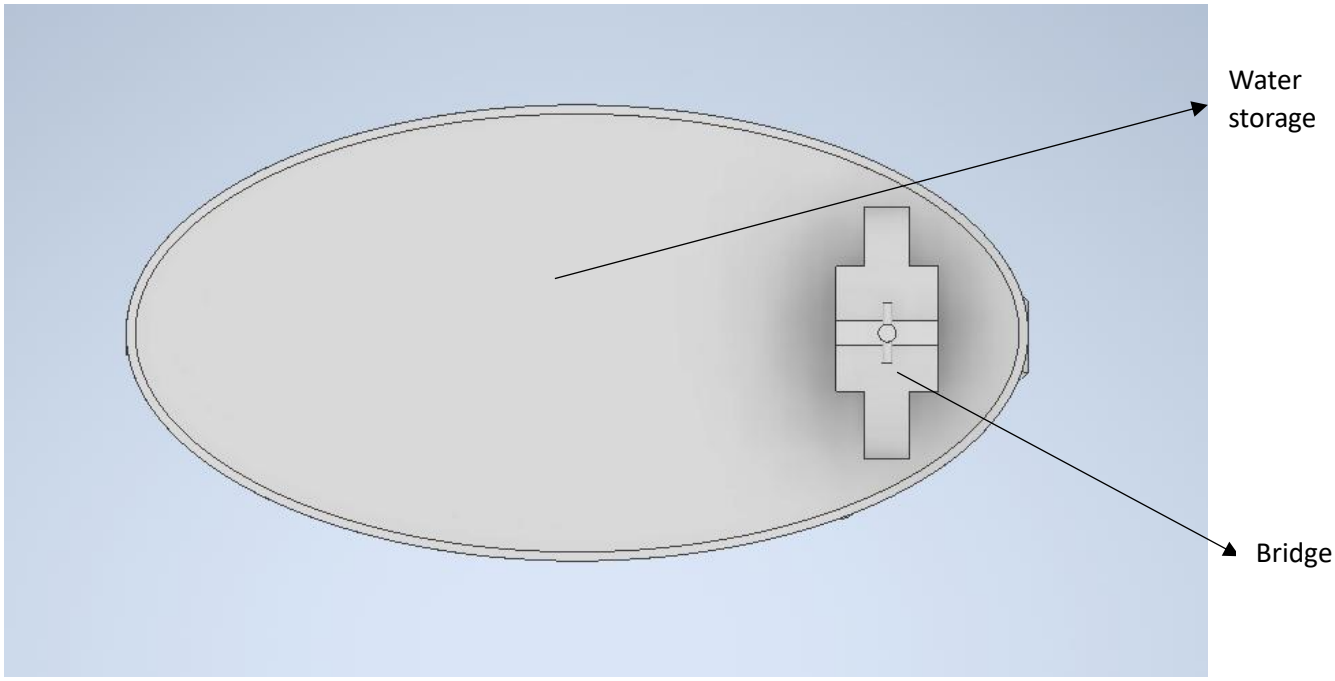


Figure 3e Top View

Part 4.4- Process

The working process of the design will be as follows: Firstly, the module is attached onto a boat. If the Solid Waste module is attached, Firstly, the gate of the Filter will be opened, the boat will move, and the water alongst with the solid waste will flow through The Filter. The waste which is solid in nature will be blocked and force to be left in The Filter when the water will be able to pass through The Filter. If the Water Purification Module was added, the water will be directed for further processing, A sensor will be placed at the end of the Filter, when the Solid Waste are cumulated to a certain level, the Collector Robotic Arm will lift and collect all the waste away from the U-Shaped Space in the Filter and in to the collecting bin. Which then will be compressed further to conserve space. If the Water Purification Module was added, the water will be directed into a water tank, which pressure will be applied to it, forces it to pass through the ultrafiltration membrane, where the larger particles such as heavy metal will be left on the other side of the membrane when water molecule will be able to pass through the membrane. In order too ensure that the water can revert back to its optimum state, the processes water will then pass to he reverse osmosis chamber to be further processed. This will endure that the water will be of the most clean and uncontaminated state.

Part 5: Conclusion

In conclusion, our design provides a cutting edge in the field as it will be able to provide a design that will be implementable in both crude oil and chemical spillage. This provides us with an advantage that allows us to decontaminate water that are being affected by crude oil or/and chemicals. More specifically heavy metals.

By implementing ultrafiltration as a pre process function we will not only able to prolong the lifespan of the reverse osmosis, we will also be able to filter out the heavy metal particle that we wish to decontaminate from the chemical affected water bodies. Hence giving a cleaner water back to the environment

By implementing Reverse osmosis as a final stage of process we will be able to eliminate the other smaller contaminant particles such as crude oil, hydrocarbons and other smaller heavy metals. This will allow us to provide a cleaner and purer water back to the water bodies.

We wish to further improve our design in the future by implementing artificial intelligence to analyse the composition of the water bodies before the contamination and hence able to add in mineral salts and other healthy components so that we can ensure that the water bodies will be able to return fully to its original state

Part 6: Appendix

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